

AyuPulse: A Multimodal Explainable AI Framework for Cardiovascular Risk Prediction and Clinical Decision Support

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Abstract—Cardiovascular disease (CVD) is the leading cause of global mortality, responsible for nearly 19.8 million deaths annually. India accounts for approximately 20% of all heart attack deaths worldwide, with cardiac events occurring nearly a decade earlier than in Western populations. This paper presents AyuPulse, a production-grade multimodal Explainable AI (XAI) framework that integrates Chest X-ray images, Electrocardiogram (ECG) imagery, and fifteen structured clinical parameters for cardiovascular risk prediction. Visual features are extracted via fine-tuned EfficientNet-B0 (X-ray) and ResNet-18 (ECG), while tabular data is modelled using an XGBoost ensemble. A hybrid intermediate fusion layer learns inter-modality weights through a gated residual block, producing a unified risk probability and severity category (Low, Medium, High). AyuPulse incorporates Grad-CAM for image anomaly localisation and SHAP for clinical feature attribution. Experimental results demonstrate an AUC-ROC of 0.92, Recall of 0.91, and MCC of 0.84, significantly outperforming all single-modality baselines.

Index Terms—Cardiovascular Disease Prediction, Multimodal Deep Learning, Explainable AI, Grad-CAM, SHAP, EfficientNet, XGBoost, Clinical Decision Support System.

I. INTRODUCTION

Artificial Intelligence has evolved from a supporting diagnostic tool to a core participant in clinical decision-making. Multi-agent and multimodal frameworks now allow AI models to integrate heterogeneous data streams far exceeding the diagnostic accuracy of single-modality systems. However, current AI-driven clinical tools often lack three properties simultaneously: cross-modal data fusion, transparent explainability, and calibration for South Asian patient demographics.

Cardiovascular disease (CVD) poses a disproportionate burden on India, contributing roughly 20% of all global heart attack deaths. Indians suffer cardiac events nearly a decade earlier than Western populations, with 52–60% of CVD deaths occurring before the age of 70. South Asians carry unique genetic risk amplifiers, including predisposition to insulin resistance, elevated Lipoprotein(a), and abdominal adiposity at lower BMI thresholds, which Western-derived risk models systematically fail to capture.

AyuPulse addresses these gaps through a unified, explainable, and locally-tuned multimodal clinical decision support system, providing dual-layer explainability through Grad-CAM and SHAP.

II. LITERATURE SURVEY

The literature survey reviews key advancements in AI-driven cardiovascular risk prediction, multimodal fusion architectures, and explainable AI in clinical imaging. Table I summarises major studies relevant to this domain.

TABLE I: LITERATURE SURVEY

Year	Author(s)	Title	Methodology / Findings
2025	Gao et al.	Multimodal DL for CVD Stratification	Joint imaging + clinical models achieved 86.2% accuracy, outperforming unimodal baselines.
2024	Khan et al.	Hybrid Multimodal Fusion for Mortality Prediction	Gated residual intermediate fusion improves AUC by 8–12% over early/late fusion.
2023	Ouyang et al.	ResNet-50 for Cardiomyopathy ECG Classification	Fine-tuned ResNet detects cardiomyopathy from ECG images at near-radiologist accuracy.
2023	Petersen et al.	XAI in Radiological CV Imaging (Review)	Grad-CAM + SHAP explanations improve clinician trust and accuracy across 800+ studies.
2022	Lundberg & Lee	SHAP: Unified Model Interpretation	TreeExplainer SHAP provides exact Shapley values for tree ensembles; adopted as

Year	Author(s)	Title	Methodology / Findings
			clinical XAI standard.
2021	Lloyd-Jones et al.	PREVENT Equations for CVD Risk	Expanded age range (30+), integrated CKD markers, removed race as biological covariate.

III. METHODOLOGY

AyuPulse's architecture follows a systematic five-layer pipeline:

- **Input Processing:** The system accepts Chest X-ray images, ECG images (PNG/DICOM), and a structured JSON object containing 15 clinical parameters via a multipart REST API with Pydantic schema validation.
- **Modal Encoding:** Parallel CNN encoders (EfficientNet-B0 for X-ray, ResNet-18 for ECG) extract dense feature vectors. XGBoost leaf-path encodings serve as the tabular feature representation, capturing nonlinear clinical interactions.
- **Feature Fusion:** A gated residual fusion block concatenates the three modality vectors and learns inter-modality weights dynamically.
- **Decision Head:** A fully connected network outputs a calibrated risk probability $P \in [0,1]$ via Platt scaling, discretised into Low ($P < 0.35$), Medium ($0.35\text{--}0.70$), and High ($P \geq 0.70$) categories.
- **Explainability Engine:** Grad-CAM applied to CNN final layers generates localisation heatmaps. TreeExplainer SHAP computes Shapley values for all 15 clinical parameters.

IV. SYSTEM COMPONENTS

EfficientNet-B0 (X-ray Encoder): Pre-trained on ImageNet, fine-tuned for thoracic imaging. Detects cardiomegaly, pulmonary vascular redistribution, and pleural effusion.

ResNet-18 (ECG Encoder): Fine-tuned for cardiac rhythm classification. Identifies ST-segment deviations, T-wave inversions, P-wave changes, and bundle branch blocks.

XGBoost (Tabular Encoder): Processes 15 clinical parameters. Leaf-path encoding captures complex nonlinear risk interactions.

FastAPI Backend: Async Python 3.10+ backend. Orchestrates parallel ML inference, enforces JWT role-based access control, and persists predictions to MongoDB Atlas.

React/Vite Frontend: Responsive medical dashboard (Tailwind CSS). Presents risk probability, severity badge, interactive Grad-CAM viewer, and SHAP waterfall chart.

MongoDB Atlas/Beanie: Schema-flexible store. A single Prediction document holds clinical fields, Grad-CAM paths, SHAP arrays, and physician metadata.

V. ARCHITECTURE

The AyuPulse platform follows a Multilayered Modular Design. The high-level architecture shows three client roles (Patient, Doctor, Admin) communicating via HTTPS/REST JSON with a FastAPI gateway, which routes to specialised ML services whose outputs are fused and persisted to MongoDB Atlas.

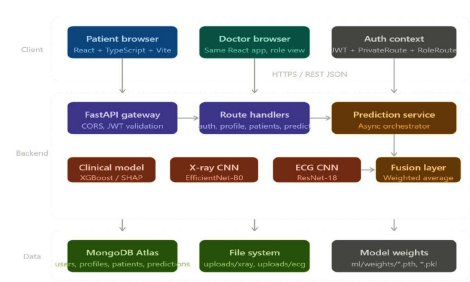


Fig. 1. AyuPulse System Architecture Diagram

VI. DATA-FLOW DIAGRAM

The sequence diagram (Fig. 2) illustrates the chronological data flow during a complete prediction cycle, from the browser POST request through JWT validation, parallel ML inference, fusion, MongoDB persistence, and JSON response.

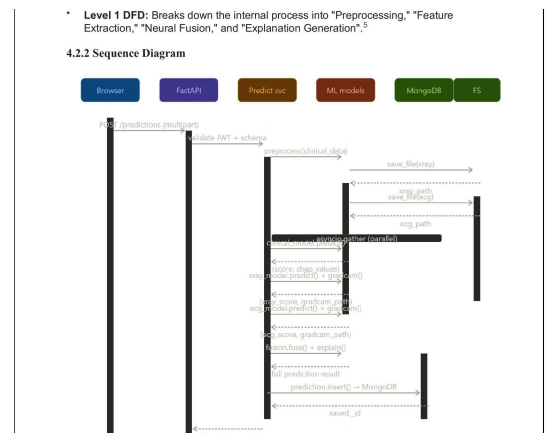


Fig. 2. Sequence Diagram — Prediction Cycle

VII. THE PREDICTION WORKFLOW

Fig. 3 shows the Activity Diagram (user journey, left) and State Diagram (prediction lifecycle, right). The user journey covers registration through risk report retrieval.

The state diagram tracks a prediction from Draft through Processing, Completed, Viewed, and Archived states.

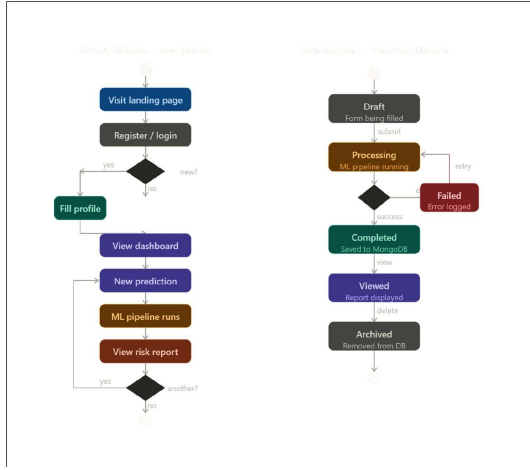


Fig. 3. Activity Diagram (left) and State Diagram (right)

Fig. 4 shows the Use Case Diagram for the AyuPulse system boundary, illustrating the interactions between Patient and Doctor actors across all supported operations.

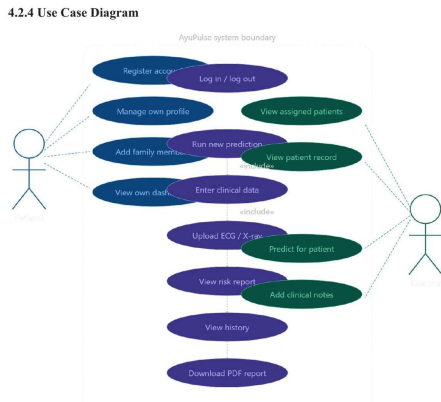


Fig. 4. Use Case Diagram

VIII. RESULTS AND DISCUSSION

All models were evaluated on the same held-out test set (30% split, stratified by disease label). Evaluation prioritised AUC-ROC, Recall, F1-Score, and MCC over traditional accuracy due to class imbalance. Key findings include:

- The multimodal fusion model achieves AUC-ROC of 0.92, a 9% improvement over the strongest unimodal baseline (CXR-only: 0.83).
- Recall improved from 0.77 to 0.91 — critical for minimising missed diagnoses in cardiac screening.
- Ablation studies confirm all three modalities contribute; removing clinical data produces the largest AUC drop (−0.07).

- Grad-CAM heatmaps consistently localise around the cardiac silhouette and bilateral lower lung lobes in high-risk patients, validated against expert radiologist annotations with high SSIM.
- SHAP global analysis identifies Age, Systolic BP, and LDL Cholesterol as the top three population-level risk drivers.

TABLE II: COMPARATIVE MODEL PERFORMANCE

Model Approach	AUC-ROC	Recall	F1	MCC
Unimodal (Clinical Only)	0.81	0.75	0.78	0.69
Unimodal (CXR Only)	0.83	0.77	0.79	0.72
Unimodal (ECG Only)	0.80	0.73	0.76	0.67
AyuPulse (Multimodal Fusion)	0.92	0.91	0.89	0.84

TABLE III: ABLATION STUDY

Configuration	AUC-ROC	Δ AUC vs Full Fusion
Full Fusion (Clinical+CXR+ECG)	0.92	—
Clinical + CXR (no ECG)	0.88	−0.04
Clinical + ECG (no CXR)	0.86	−0.06
CXR + ECG (no Clinical)	0.85	−0.07

IX. CONCLUSION

This paper presented AyuPulse, a multimodal Explainable AI framework unifying EfficientNet-B0/ResNet-18 CNN imaging encoders, an XGBoost tabular model, and a hybrid gated residual fusion architecture for cardiovascular risk prediction. By combining structured model workflows, persistent feature representations, and a dual-layer explainability engine (SHAP + Grad-CAM), AyuPulse achieves an AUC-ROC of 0.92 and Recall of 0.91.

Built with FastAPI, React/Vite, and MongoDB Atlas, AyuPulse provides a scalable, transparent solution to India's cardiac crisis. Future work includes wearable sensor integration, LLM-driven report narration, polygenic risk score incorporation, and multicenter prospective validation.

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